LESSON 9.10

SIMILAR TRIANGLES AND DILATIONS IN THE REAL WORLD

LESSON INTRODUCTION

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

LEARNING TARGETS

- I can solve real-world problems that involve similar triangles.

BENCHMARK COHERENCE

BUILDING ON

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

WORKING TOWARD

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

ESSENTIAL QUESTIONS

- How are similar triangles and dilations of triangles related?
- How can you use similar triangles and dilations to solve real-world problems?

LESSON NARRATIVE

In this culminating lesson, students apply their prior knowledge of dilations, the Pythagorean Theorem, and area and perimeter of triangles to solve real-world problems involving similar triangles. By applying prior knowledge students have the opportunity to further solidify their learning.

COMPONENT SUMMARY

9.10.1 WARM-UP *(~5 MIN)*

Review right triangles

9.10.2 GUIDED PRACTICE *(15 MIN)*

Use the Pythagorean Theorem in real-world situations

9.10.3 COLLABORATION *(20 MIN)*

Use dilations to solve real-world problems with triangles

9.10.4 WRAP-UP *(~5 MIN)*

Assess your learning

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RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- Calculators
- (Optional) Highlighters or Colored Pencils
- (Optional) Patty Paper

ADDITIONAL PREPARATION

- Consider creating cutouts on cardstock of the triangles in the real-world examples.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to see the similar triangles when the image and preimage share angle or sides.
- Students may struggle determining which formula (area, perimeter, Pythagorean Theorem, scale factor of dilation) to use on which problem.

THINGS TO CONSIDER

- Consider allowing students to use calculator to determine the approximation of a square root. In problems that require several steps students should not round until the final step. Rounding mid-step could significantly change the answer. Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common careless errors that many students make.
- Consider providing strategies for students when working with overlapping triangles. Such strategies include highlighting the triangles using different colors.

LITERACY AND LANGUAGE ROUTINES

MLR 6: Three Reads

Use this routine to ensure that students accurately comprehend the dense text in the problems in 9.10.2 Guided Practice and 9.10.3 Collaboration. For the first read, someone should read the text aloud. Provide definitions and examples for any non-mathematical terms students are not familiar with. Then, prompt students to focus on what the question is asking them to find, what information they are given, and to develop a problem-solving plan.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

In 9.10.2 Guided Practice and 9.10.3 Collaboration students have opportunities to solve real-world problems involving dilations. Discussing applications of the content they are learning about will help students gain appreciation for the utility of mathematics.

DIFFERENTIATE INSTRUCTION

The images of the dilations in 9.10.2 Guided Practice and 9.10.3 Collaboration provide a visual entry-point. Prompt students to color code corresponding sides and angles. The discussions in 9.10.3 Collaboration provide an auditory entry-point. Consider providing students with cardstock cutouts so they can physically manipulate the triangles in the real world examples.

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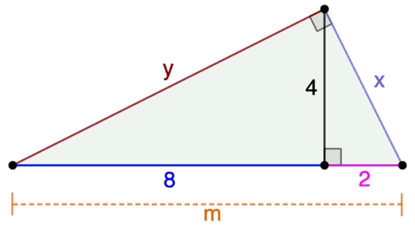
9.10.1 WARM-UP: RELATIONSHIPS IN TRIANGLES

TEACHER MOVES

Students discovered the Pythagorean Theorem in Unit 5. The Warm-Up refreshes students' memory of when and how to use the Pythagorean Theorem. Consider prompting students to compare their equations from question 1b with a partner and collaboratively generate all three equations involving the Pythagorean Theorem.

9.10.1 Warm-Up: Relationships in Triangles

1. A figure is shown. Side lengths and distances, in units, are labeled using variables and known quantities.



   a. How many right triangles do you see in the figure?
   3

   b. Write at least two equations that relate some of the side lengths or distances marked in the figure.
   Answers will vary.
   $x^2 + y^2 = 100$
   $2^2 + 4^2 = x^2$

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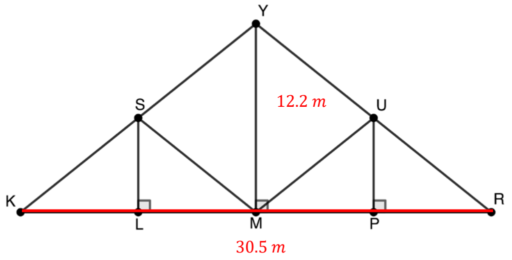
9.10.2 Guided Practice: Dilations and Similar Triangles in the Real World

The Waddell "A" Truss Bridge was built in 1898 and connects Parkville, MO to Trimble, MO. Built using steel rods, the bridge was originally used by the railway but is currently a pedestrian bridge.

1. A scale drawing of the bridge is shown.



The total span of the bridge is 30.5 meters (m), and the height at its highest point is 12.2 m. The span of a bridge is the length of the bridge.

a. Label the given measurements on the figure.

b. Complete each statement using the given information.

The span of the bridge is represented by line segment $\underline{KR}$.

9.10.2 GUIDED PRACTICE: DILATIONS AND SIMILAR TRIANGLES IN THE REAL WORLD

TEACHER MOVES

In this culminating lesson students incorporate several different concepts to solve problems. Allow students to have some productive struggle in recalling information. This struggle will solidify their understanding of these concepts. Allow students to use their workbooks and other resources to recall the content. As you guide students through the questions consider where longer time may be needed for student recall. Give ample time for all students to find information to help them.

QUESTIONING

Consider inviting an engineer to join the class (virtually or in-person) to share how bridges are designed and why they often contain triangles.

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9.10.2 GUIDED PRACTICE: DILATIONS AND SIMILAR TRIANGLES IN THE REAL WORLD (CONTINUED)

TEACHER MOVES

Notice and Wonder

Ask students what they notice and wonder about the image. Avoid evaluating the correctness of students' statements as the it may discourage students from sharing in the future. Prompt student-to-student interaction by asking students if they agree/disagree and why.

DIFFERENTIATE INSTRUCTION

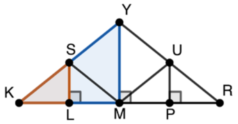
Encourage students to use highlighters or colored pencils to color code the corresponding sides and angles in the triangles. Consider providing cardstock cutouts of the triangles so students can physically manipulate the triangles.

TEACHER MOVES

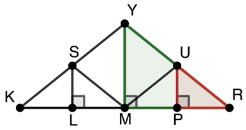
Students often struggle with overlapping triangles. The coloring of the triangles is to support students in distinguishing between the two triangles but some students may be visually confused. Use students' notices and wonders to help direct supporting student visualization of the overlapping triangles. Consider different strategies to help students distinguish the different triangles. Strategies include using a highlighter and drawing the two triangles separately.

The height of the bridge is represented by line segment MY.

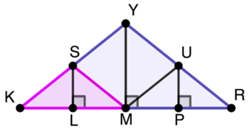
The images and statements below give additional information about the bridge design.



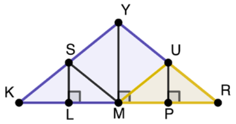
ΔKLS is a dilation of ΔKMY using a scale factor of $\frac{1}{2}$ centered at point K.



ΔRPU is a dilation of ΔRMY using a scale factor of $\frac{1}{2}$ centered at point R.



ΔRKY is a dilation of ΔMKS using a scale factor of 2 centered at point K.



ΔRKY is a dilation of ΔRMU using a scale factor of 2 centered at point R.

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c. Use the Pythagorean Theorem, properties of similar triangles, and other reasoning to find the lengths of the line segments. Round to the nearest thousandth of a meter.

Line Segment	Work/Reasoning	Length (m)
$\overline{KR}$	Given	30.5
$\overline{YM}$	Given	12.2
$\overline{SL}$	*SL is $\frac{1}{2}$ of YM*	6.1
$\overline{UP}$	*UP is $\frac{1}{2}$ of YM*	6.1
$\overline{YK}$	*MK is $\frac{1}{2}$ of KR, so 15.25* *Using Pythagorean Theorem* $15.25^2 + 12.2^2 = YK^2$	19.530
$\overline{YR}$	*YR is the same length as YK*	19.530
$\overline{SM}$	*SM is $\frac{1}{2}$ of YR*	9.765
$\overline{UM}$	*UM is $\frac{1}{2}$ of YK*	9.765

d. How many total meters of steel were used to create one side of this bridge?
113.49 *meters*

9.10.2 GUIDED PRACTICE: DILATIONS & SIMILAR TRIANGLES IN THE REAL WORLD (CONTINUED)

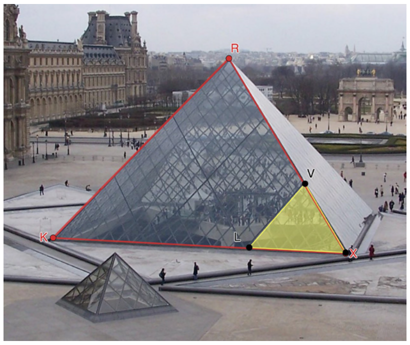
TEACHER MOVES

Consider having students complete the table as much as they can on their own. Then, allow students some time to confer with a peer on the reasoning for measurements they cannot figure out on their own. Alternatively, have students complete the table and then to verify their reasoning and answers with their partners.

9.10.3 Collaboration: Dilations & Similar Triangles in the Real World

The Louvre in Paris is a famous museum that houses artwork like the *Mona Lisa*. Though the Louvre building was originally built in 1190 to serve as a fortress, four glass pyramids were built in the courtyard of the Louvre in 1989. These pyramids – one large one in the center, with three smaller ones along the sides – are now the defining structures of the Louvre Museum.

Each panel of the pyramids is built from numerous pieces of glass. The diagram shows ΔRXK and ΔVXL.



9.10.3 COLLABORATION: DILATIONS & SIMILAR TRIANGLES IN THE REAL WORLD

TEACHER MOVES

Three Reads

Use this routine to ensure that students accurately comprehend the dense text in the problems. For the first read, someone should read the text aloud. Provide definitions and examples for any non-mathematical terms students are not familiar with. Then, prompt students to focus on what the question is asking them to find, what information they are given, and to develop a problem-solving plan.

DIFFERENTIATE INSTRUCTION

Consider inviting an architect to join class (virtually or in-person) to share information about why and how they use triangles in their designs.

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9.10.3 COLLABORATION: DILATIONS & SIMILAR TRIANGLES IN THE REAL WORLD (CONTINUED)

ENRICHMENT

- What are the corresponding sides and angles in triangles VXL and RXK?
- What information is given? What are you asked to find?
- How can you use the information you are given to determine what you are asked to find?

DIFFERENTIATE INSTRUCTION

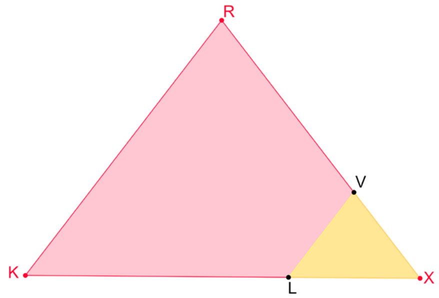
The discussions during this collaborative activity provide an auditory entry-point. Encourage students to color code corresponding sides and angles.

TEACHER MOVES

In the case of successive questions, it is natural for students to utilize the answer from one to another. Use this opportunity to discuss with students why using a rounded value could result in an incorrect answer. Discuss with students how much 0.02 feet (2.4 inches) could make a difference in the building of the panel.

Work with your partner to complete the following.

1. A scale drawing of one side panel of the large pyramid, ΔRXK, is shown, including ΔVXL. $RK = 106\frac{1}{2}$ feet (ft) and $LX = 37\frac{1}{3}$ ft.



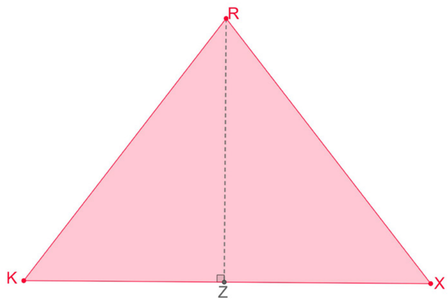
a. In this design, the full panel, ΔRXK can be determined by dilating the piece, ΔVXL. Explain which point is the center of dilation.

The center of dilation is point X.

b. Given the scale factor of dilation is 3, find the side lengths of the full panel, ΔRXK.

Side	Work	Length (in feet)
RK	*Given*	$106\frac{1}{2}$ *feet*
RX	$RX = RK$	$106\frac{1}{2}$ *feet*
KX	$37\frac{1}{3} \times 3$	112 *feet*

c. Label the side lengths of ΔRXK.



d. Use the Pythagorean Theorem to find the height of the panel. Round your answer to the nearest thousandth of a foot.
$RZ^2 + 56^2 = 106.5^2$
$RZ^2 + 3{,}136 = 11{,}342.25$
$RZ = 90.588$

e. Find the area of the panel, in square feet. Round your answer to the nearest hundredth of a square foot.
$A = \frac{1}{2}(RZ)(KX)$
5,072.948 *if using unrounded value.*
5,072.928 *if using rounded value.*

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9.10.4 Wrap-Up: Reflection

1. Complete the table.
Answers will vary.

Mindset	Choose the best description.
Did I use the partner discussions learning opportunities?	○ Never ○ Sometimes ○ Usually ○ Always
Did I work on problems that challenged me?	○ Never ○ Sometimes ○ Usually ○ Always
Did I use strategies to 'un-stick' myself when I found the problems difficult?	○ Never ○ Sometimes ○ Usually ○ Always
Did I put as much effort as I possibly could into the problems?	○ Never ○ Sometimes ○ Usually ○ Always

9.10.4 WRAP-UP: REFLECTION

TEACHER MOVES

Consider how this reflection indicates how students are progressing with a growth mindset. For students who are timid about trying problems or making mistakes look for ways to model a growth mindset as you work with these students.

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